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Predicting Exercises Based on User-Selected Variance

Abstract

An exercise recommendation system determines workout plans for users. The exercise recommendation system receives a profile of a user and a level of variance selected by the user. The profile includes a history of exercises the user has performed, available gym equipment, and exercise goals. The exercise recommendation system inputs the profile to a machine learning model configured to rank a set of exercises for a user to perform. The exercise recommendation system modifies the ranking of exercises based on the level of variance selected by the user. Modification of the ranking is greater for a first level of variance than for a second level of variance less than the first level of variance. The exercise recommendation system generates a workout plan for display within a user interface to the user based on the modified ranking.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Application No.: 17/233,238, filed on Apr. 16, 2021, which is incorporated herein by reference for all purposes.

BACKGROUND

[0002] This invention relates generally to determining a workout plan for a user and, more particularly, to using machine learning to determine exercises, weights, and capabilities for users.

[0003] When working out, users may follow different workout plans from one another. For example, one user may want to focus on arm exercises one day and leg exercises the next day with a goal to strengthen their muscles each day. In another example, a user may prefer that the exercises in their workout be the same every week, while another user may prefer to perform a greater variety of exercises when working out. However, current systems do not allow a user to specify that they would like variety in their workout while accounting for the user's goals and other preferences in determining a workout for the user.

[0004] Further, some users may want to increase their capabilities when they work out, such as by increasing a weight that they use or number of repetitions they perform of an exercise over time. However, current systems lack the ability to predict a user's capability based on their previous exercises and how recently they worked out. Thus, a system that determines a user's capability for exercises is necessary.

SUMMARY

[0005] The following disclosure describes an exercise recommendation system that selects exercises for a user to perform. In particular, the exercise recommendation system receives a profile of a user and a level of variance selected by the user. The profile may include a history of exercises the user has performed, available gym equipment, and one or more exercise goals. The exercise recommendation system inputs the profile of the user to an exercise selection model, which is configured to rank a set of exercises for a user to perform based on the history of exercises the user has performed, the available gym equipment, and the one or more exercise goals. The exercise recommendation system modifies the ranking of exercises based on the level of variance selected by the user. Modification of the ranking of exercises is greater for a first level of variance than for a second level of variance less than the first level of variance. The exercise recommendation system generates a workout plan for display within an interface to a user based on the modified ranking.

[0006] In some embodiments, the exercise recommendation system also selects target weights for users. In particular, the exercise recommendation system accesses a plurality of exercise pairs. Each exercise pair is associated with a weight and labeled with performance statistics of users who have completed the exercises of the exercise pair. The exercise recommendation system trains a machine learning model using the plurality of exercise pairs. The machine learning model is configured to determine a weight to recommend to a user for a first exercise based on performance statistics of user associated with one or more second exercises, where each second exercise is within an exercise pair with the first exercise. The exercise recommendation system retrieves performance statistics of a target user including a weight for each of one or more exercises previously performed by the target user. The exercise recommendation system applies the machine learning model to the retrieved performance statistics to determine, for a target exercise, a target weight to recommend to the target user for the target exercise. The exercise recommendation system modifies an interface displayed by a client device of the target user to include the target

weight.

[0007] In further embodiments, the exercise recommendation system determines current capabilities of users. In particular, the exercise recommendation system accesses an exercise history for a user. The exercise history comprises an exercise performed by the user and a capability of the user each time the user performed the exercise. The exercise recommendation system partitions the exercise history into a plurality of time periods. For each time period, the exercise recommendation system computes an aggregate capability of the user for the exercise based on the capabilities of the user in performing the exercise during the time period. The exercise recommendation system calculates a moving average capability of the user for the exercise based on the aggregate capabilities. The exercise recommendation system determines a current capability of the user for the exercise based on the moving average capability. The current capability of the user for the exercise is discounted at least in part based on how recently the user performed the exercise.

[0008] The features and advantages described in the specification are not all inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes, and may not have been selected to delineate or circumscribe the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 illustrates a system environment for an exercise recommendation system, according to one embodiment.

[0010] FIG. 2 is a high-level block diagram of an exercise recommendation system, according to one embodiment.

[0011] FIG. 3 is a flowchart illustrating a process for generating a workout plan, according to one embodiment.

[0012] FIG. 4 is a flowchart illustrating a process for modifying an interface to display a target weight, according to one embodiment.

[0013] FIG. 5 is a flowchart illustrating a process for determining a current capability of a user, according to one embodiment.

[0014] The figures depict various embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

[0015] FIG. 1 illustrates a system environment for an exercise recommendation system **100**, according to one embodiment. The exercise recommendation system **100** is connected to a number of client devices **120** used by users of the exercise recommendation system **100** via a network **110**. These various components are now described in additional detail.

[0016] The client devices **120** are computing devices such as smart phones, laptop computers, desktop computers, or any other device that can communicate with the exercise recommendation system **100** via the network **110**. The client devices **120** may provide a number of applications, which may require user authentication before a user can use the applications, and the client devices **120** may interact with the exercise recommendation system **100** via an application. Though two client devices **120** are shown in FIG. 1, any number of client devices **120** may be connected to the exercise recommendation system **100** in other embodiments.

[0017] The network **110** connects the client devices **120** to the exercise recommendation system

100, which is further described in relation to FIG. 2. The network **110** may be any suitable communications network for data transmission. In an embodiment such as that illustrated in FIG. 1, the network **110** uses standard communications technologies and/or protocols and can include the Internet. In another embodiment, the network **110** use custom and/or dedicated data communications technologies.

[0018] FIG. 2 is a high-level block diagram of an exercise recommendation system **100**, according to one embodiment. The exercise recommendation system **100** operates in real-time to determine exercises, target weights, target repetitions, capabilities, and the like for users. The exercise recommendation system **100** includes a workout plan module **200**, an exercise selection model **210**, a recommendation module **220**, a weight recommendation model **230**, a capability module **240**, a training module **250**, a user interface module **260**, a user profile datastore **270**, an exercise datastore **280**, and a training datastore **290**. In some embodiments, the exercise recommendation system **100** may include more modules or models than shown in FIG. 2 or one or more of the modules and models shown in FIG. 2 may be combined within the exercise recommendation system **100**.

[0019] The workout plan module **200** determines workout plans for users. A workout plan includes one or more exercises for a user to complete during a workout (e.g., a continuous time period during which a user performs exercises). An exercise is an activity that a user performs to improve their physical fitness, and exercises are stored in the exercise datastore. Examples of exercises may include squats, leg lifts, jumping jacks, and the like. Each exercise is associated with exercise information, which may include an exercise type (e.g., cardio, bodyweight, weighted, warm-up, timed intervals, etc.), a list of users who have performed the exercise, and exercise requirements. Exercise requirements describe equipment needed to perform an exercise or physical ailments that may preclude a user from performing the exercise. Exercises and exercise information is stored in the exercise datastore **280**.

[0020] The workout plan module **200** receives a request for a workout plan from the user interface module **260**. The request can include a level of variance selected by the user via a user interface, as described in relation to the user interface module **260** below. The level of variance represents how much variety in exercise type, exercise, workout length, and focused muscle group the user would like in a workout. For instance, a high level of variance (e.g., 80%) indicates that a user prefers to perform a variety of exercises each workout whereas a low level of variance (e.g., 25%) indicates that a user prefer to perform the same exercises each workout. In some embodiments, the workout plan module **200** may use a default level of variance as the selected level of variance if the user has not explicitly selected a level of variance via the request. In other embodiments, the workout plan module **200** may specify a level of variance of the user based on a history of exercises performed by the user stored in the user profile datastore **270**, as described below. For example, if the user has performed a certain number of workouts or number of exercises, the workout plan module **200** may automatically adjust level of variance to correspond to the user's history.

[0021] The workout plan module **200** retrieves a profile of the user from the user profile datastore **270**. The user profile datastore **270** stores profiles of users, where each profile includes a current height, weight (which may be stored in pounds or kilograms), age, muscle mass, and level of fitness experience (e.g. beginner, intermediate, advanced, etc.) of a user. The profile may also include historical weights of the user, historical muscle masses of the user, gym equipment available to the user, one or more exercise goals selected by the user (e.g., number of calories to burn, desired weight, desired muscle mass, desired strength per exercise, desired type, format, or style of workout, etc.), ratings of exercises previously performed by the user, one or more muscle groups the user would like to target during workouts, and any exercise restrictions indicated by the user (e.g., the user has knee issues and cannot squat), and an exercise history of exercises previously performed by the user during one or more workouts. Each exercise in the exercise history may be associated with performance statistics of the user when the user completed the exercise and a recency (e.g., how recently the exercise was performed by the user). For instance,

the performance statistics may include a weight the user used for the exercise (stored as pounds or kilograms), a number of sets the user completed of the exercise, a number of repetitions per set the user completed for the exercise, and a time to complete the exercise. The performance statistics may also include the user's current weight and muscle mass. Further, the exercise history may be associated with an exercise list for the user which includes exercises that match the user's profile (e.g., exercises that the user has equipment and ability to perform). The exercise list may alternatively be stored in the exercise datastore **280**.

[0022] Each profile may also include a capability of a user. The capability is a maximum amount of weight that the user can use to do one repetition of an exercise. For example, the user's capability may be 150 pounds of weight doing one bench-press. In some embodiments, capability may include weight capability and repetition capability for a particular exercise (e.g., bench-press). A repetition capability may be a maximum number of repetitions the user can do of the exercise in a single set. For example, the user's weight capability may be using 150 pounds of weight to do one bench-press, and the user's repetition capability may be doing 20 bench-presses with 50 pounds of weight. In some embodiments, the repetition capability may be only used for bodyweight exercises that do not require the user to use extra weight. or a number of a weighted exercise in a set using a base weight. The capability (e.g., weight capability and repetition capability) of the user for an exercise may be stored in the exercise datastore **280** in association with the exercise in the user's exercise list. The capability module **240** may update the capability of the user for each exercise as the user completes exercises, as further described below.

[0023] The workout plan module **200** inputs the profile of the user to the exercise selection model **210**. The exercise selection model **210** is a machine learning model configured to rank a set of exercises for a user to perform based on a profile of a user, including the exercise history that the user has performed (e.g., as stored in for the user's profile), gym equipment available to the user, and one or more exercise goals. The exercise selection model **210** may also be configured to rank the exercises based on exercise metadata stored in the exercise datastore **280**. The exercise metadata may include a score for each exercise indicating an efficacy of the exercise and/or whether the exercise targets primary muscle groups (e.g., quadriceps, shoulders, back, etc.) or accessory muscle groups (e.g., abductors, forearms, calves, etc.). The exercise selection model **210** may be a classifier, decision tree, regression model, neural network, random forest model, or the like. In some embodiments, the exercise selection model **210** may be a statistical model. The exercise selection model **210** is trained by the training module **250**, as described below.

[0024] The workout plan module **200** receives, from the exercise selection model **210**, scores of exercises of the exercise datastore **280**, which form a ranking of the exercises based on the scores. In some embodiments, the workout plan module **200** may store an exercise list specific to each user in the exercise datastore **280** such that the exercise selection model **210** may score and rank the exercises from the user's exercise list. In these embodiments, the exercise list may include exercises that the user has equipment for or that align with the exercise goals set by the user. The workout plan module **200** adjusts the ranking based on the profile of the user who submitted the request. For example, if the user's profile indicates that the user has a goal to improve their quad strength, the workout plan module **200** may boost scores for exercises that focus on strengthening quads by 1.5 times (or another multiplicative factor). In another example, if the user's profile indicates that they do not own dumbbells, the workout plan module **200** may reduce the scores of exercises that require dumbbells to zero or remove the exercises from the ranking.

[0025] The workout plan module **200** may also adjust the scores in the ranking based on the selected level of variance. In some embodiments, the workout plan module **200** may alter the ranking of exercises completed by the user within a threshold amount of time by the selected level of variance. For example, for a selected level of variance of 50%, the workout plan module **200** may randomly select 50% of the exercises the user most recently performed (e.g., within a threshold amount of item) and increase their rankings by 5 places. The workout plan module **200**

may additionally decrease the rankings of the other 50% of exercises most recently performed by 5 places. The workout plan module **200** may reduce the scores of exercises recently performed by the user, exercises with low ratings by the user (e.g., on a scale from 1 to 5, a 3 or less), or exercises that require equipment the user does not have by a set amount (e.g., subtracting one from each score) or to zero. Alternatively, the workout plan module **200** may reduce the scores of exercises that the user does not have equipment for to zero or may alter the score by the level of variance (e.g., multiplying a subset of the scores by the level of variance).

[0026] The workout plan module **200** selects one or more of the exercises for the user's workout plan based on the ranking. For instance, the workout plan module **200** may select the top ten ranked (or another threshold number of) exercises from the exercise list send those to the user interface module **260** for display to the user as the user's workout plan. In some embodiments, the workout plan module **200** may select the top-ranked exercise to send to the user interface module **260** for display to the user and may use the exercise selection model **210** to re-score and re-rank the exercises from the exercise list to select an next exercise for the user to perform upon receiving an indication from the user interface module **260**. The workout plan module **200** may repeat this process until the workout plan module **200** has selected a threshold number of exercises for the user to perform. In other embodiments, the workout plan module **200** may select the top-ranked exercise and subsequently use the exercise selection model **210** to re-score and re-rank the exercises from the exercise list to select a next exercise. The workout plan module **200** may repeat this process of re-scoring, re-ranking, and selecting until the workout plan module **200** has selected a threshold number of exercises for the workout plan, which the workout plan module **200** sends to the user interface module **260** to display to the user as the workout plan.

[0027] Further, in some embodiments, the workout plan module **200** selects a subset of exercises to send for the workout plan based on the level of variance. For instance, the workout plan module **200** may select more exercises for a high level of variance than for a low level of variance. In these embodiments, the workout plan module **200** may randomly select exercises from the subset to send to the user interface module **260** for the workout plan upon receiving a request from the user interface module **260**. Thus, the user may see a greater variety of exercises for a higher level of variance than a lower level of variance.

[0028] For each selected exercise, the workout plan module **200** request a recommended weight, number of repetitions, and number of sets (henceforth referred to collectively as recommendations) from the recommendation module **220**. For example, the exercise "bicep curls" may be associated with a weight of 15 pounds per arm, 10 curls, and 3 sets for in the user's workout plan. To complete the exercise, the user could to use 15-pound dumbbells to do 10 bicep curls on each arm 3 times. The recommendation module **220** may determine a weight, number of repetitions, and number of sets to recommend to the user based on the user's performance statistics.

[0029] The recommendation module **220** may receive a request for the one or more of the recommendations for a user and an exercise from the workout plan module **200**. Alternatively, the recommendation module **220** may receive a request for the one or more recommendations and the exercise from the user interface module **260**. The recommendation module **220** retrieves performance statistics of the user who submitted the request from the user profile datastore **270** and uses the performance statistics to determine recommendations for the user. In some embodiments, if the user has previous performed the exercise, the recommendation module **220** may base the recommendations on the most recent performance statistics related to the exercise in the exercise in the history. In other embodiments, if the user has not performed the exercise, the recommendation module **220** may apply a ratio based on an exercise pair including the exercise and an exercise the user has performed to determine recommendations. Alternatively, the recommendation module **220** request a capability of the user from the capability module **240**, which is described below, and use the capability of the user to determine recommendations for the user.

[0030] Further, the recommendation module **220** may input the performance statistics of the user to

the weight recommendation model **230** to determine the recommendations. The weight recommendation model **230** is a machine learning model configured to determine a weight to recommend to a user for an exercise based on performance statistics of the user associated with other exercises. The weight recommendation model **230** may additionally receive the exercise as an input or may output a weight for each exercise in the exercise datastore **280**. In some embodiments, the weight recommendation model **230** may also determine the number of repetitions and number of sets of the exercise to recommend to the user. The weight recommendation model **230** may be a classifier, decision tree, regression model, neural network, random forest model, or the like. The weight recommendation model **230** is trained by the training module **250**, as described below.

[0031] In some embodiments, the recommendation module **220** receives recommendations directly from the weight recommendation model **230**. In some embodiments, the recommendation module **220** may receive a score for each weight in a list of weights. For example, the list of weights may include weights from zero pounds to **200** pounds (or another upper limit of weight) at five-pound increments. The recommendation module **220** may access the user's profile and reduce the scores of weights that the user does not own to zero. The recommendation module may also reduce the scores of weights that do not align with the user's exercise goals or restrictions. For example, if the user's goal is "aerobic exercise," the recommendation module **220** may reduce the scores of weights that are over a third of the user's personal weight. The recommendation module **220** ranks the weights based on the scores and selects the highest-ranked weight to recommend to the user. The recommendation module **220** may similarly receive scores for one or more number of repetitions and/or number of sets from the weight recommendation model **230** and, after reducing one or more scores based on the user's profile, rank and select a number of repetitions and/or number of sets based on the ranking. The recommendation module **220** may send the recommendations to the user interface module **260** for display to a user or may send the selections to the workout plan module **200**.

[0032] The workout plan module **200** receives the recommendations from the recommendation module **220** for each selected weight. The workout plan module **200** may determine an order for the exercises based on the ranking of exercises or may shuffle the exercises into an order randomly. The workout plan module **200** stores the order of exercises each associated with the received recommendations as a workout plan for the user in the exercise datastore **280**. The workout plan may be stored in association with a current time/date and current performance statistics of the user in the exercise datastore **280**. The workout plan module **200** send the workout plan to the user interface module **260** for display to the user. In some embodiments, the workout plan module **200** may receive indications from the user interface module **260** for each of the exercises in the workout plan indicating if the user completed the exercise using the recommendations or if the user modified the recommendations (.e.g., performed another exercise, used a different weight, requested a different exercise, etc.). The workout plan module **200** stores a record of the indications in the exercise datastore **280** as part of the exercise history for the user.

[0033] In some embodiments, the user's exercise goals may indicate that they would like to perform exercises at their capability or the user may request their capability for an exercise via the user interface module **260**. In these embodiments, the capability module **240** may receive the request from the user interface module **260** or may receive an indication from the user interface module **260** that the user inputted an exercise goal of performing at their capability. The capability module **240** may determine a current capability of the user in performing one or more exercises in response to receiving the request or indication.

[0034] In some embodiments, if the user has not performed the exercise, the capability module **240** may input the user's profile to a random forest model to determine the current capability of the user based on other user's capabilities. In other embodiments, the capability module **240** may use the capability of the user in performing other exercises to determine the user's current capability. For example, if the user has recently performed a bench-press and wants to know their current

capability for a lateral press, the capability module **240** access a ratio for the two exercises (e.g., the bench-press and lateral press) stored in the exercise datastore **280**. The capability module **240** applies the ratio to the capability of the user in performing the bench-press to determine the capability of the user in performing the lateral press. In further embodiments, the capability module **240** may calculate a moving average capability based on the user's exercise history if the user has performed the exercise before.

[0035] In particular, the capability module **240** accesses the exercise history of the user from the user profile datastore **270** to retrieve exercises performed by the user and a capability of the user each time the user performed the exercise. For example, the capability module **240** may retrieve exercise information indicating that the user has performed one bench-press using 150 pounds one week ago and one bench-press using 155 pounds one day ago. In some embodiments, the capability module **20** may only access the exercise history for one exercise specified by the request from the user and determine the user's capability for that exercise. Further, the capability module **240** may only access exercises of the exercise history performed within a recent time period (e.g., the last 180 days).

[0036] The capability module **240** partitions the exercise history into time periods. For instance, the capability module **240** may group exercises in the exercise history into one week time periods, one month time periods, or any other grouping of time. For each time period, the capability module **240** computes an aggregate capability of the user for each exercise based on the capability of the user when performing the exercise during the time period. The aggregate capability may be a sum, average, or another combination of the capability of the user for the exercise during the time period. For example, if the exercise history indicates that the user performed one bench-press using 150 pounds each day during the time period, the aggregate capability for the time period may be 150 pounds. Further, the capability module **240** may only average the capability for the exercise on days when the user actually performed the exercise and not include days where the user did not perform the exercise in computing the aggregate capability. The capability module **240** calculates a weighted moving average capability of the user based on the aggregate capabilities of the time period for each exercise. For instance, the capability module **240** may weight the aggregate capability of the most recent time period highest and the aggregate capability of the oldest time period the least.

[0037] The capability module **240** determines a current capability of the user for each exercise based on the moving average capability and a current time. In particular, the capability module **240** may use the moving average capability as the current capability or may apply exponential smoothing to the moving average capability to determine the current capability. However, if the current time is a threshold amount of time past the last time the user performed the exercise, the capability module **240** may discount the moving average capability based on how recently the user last performed the exercise. For instance, the capability module **240** may decay the moving average capability and may select a current capability from the decayed portion of the moving average capability if the user is performing the exercise more than a week (or another threshold time period) since the last performance of the exercise. The capability module **240** sends the selected current capability to the user interface module **260** for display and may store the current capability in the user's profile in the user profile datastore **270**.

[0038] The training module **250** trains the exercise selection model **210** to rank a set of exercises of a user to perform based on the exercise history of the user, the gym equipment available to the user, and one or more exercise goals set by the user. In particular, the training module **250** may access the user profile datastore **270** to retrieve profiles of users. The training module **250** labels the exercise history of each user with the user's available exercise equipment, one or more exercise goals, and other aspects from the user's profile. The training module **250** stores the labeled exercise histories in the training datastore **290** as training data for the exercise selection model **210**. In some embodiments, the training data may include training data for the exercise selection model **210** that

was manually labeled by an external operator rather than the training module **250**.

[0039] The training module **250** trains the exercise selection model **210** on the training data such that the exercise selection model **210** is configured to rank the set of exercises in the exercise datastore **280** upon receiving input of the user's profile, including the exercise history, available gym equipment, and exercise goals. In some embodiments, the exercise selection model **210** may comprise multiple machine learning models each trained for a particular user. In these embodiments, the training module **250** may train each machine learning model on training data specific to a user, and the workout plan module **200** may apply the machine learning model for the user. The training module **250** may train the exercise selection model **210** periodically as new users input profiles to the exercise recommendation system **100**.

[0040] The training module **250** also trains the weight recommendation model **230** to determine a weight to recommend to a user for an exercise. The training module **250** accesses the exercise datastore **280** to retrieve a plurality of exercise pairs, which each includes two exercises. The training module **250** accesses the user profile datastore **270** to retrieve exercise histories of users and labels each exercise pair with performance statistics of users who completed both exercises of the pair. The performance statistics of the user include any weights used for exercises, number of repetitions of the exercises done by the user, and number of sets of the exercises done by the user, and, in some embodiments, each exercise in an exercise pair is associated with a particular weight and the exercise pair is labeled with the performance statistics of users who used the associated weight when performing the exercise. The training module **250** stores the labeled exercise pairs in the training datastore **290** as training data for the weight recommendation model **230**.

[0041] The training module **250** trains the weight recommendation model **230** on the training data such that the weight recommendation model **230** is configured recommend a weight to a user for an exercise upon receiving performance statistics and the exercise. In some embodiments, weight recommendation model **230** may be further configured to determine a number of repetitions, number of sets, rest time between sets, interval time for each set, distance for an exercise (e.g., a mile, 100 meters, etc.), elevation, resistance, and/or resistance band tension to recommend to the user for the exercise. The training module **250** may train the weight recommendation model **230** periodically as more users complete exercises using the exercise recommendation system **100**.

[0042] The user interface module **260** generates and transmits a user interface (or “interface”) to one or more client devices **120** of users of the exercise recommendation system **100**. The user interface may display exercises of workout plans determined by the workout plan module **200**, recommended weights (or number of sets or repetitions) determined by the recommendation module **220**, and capabilities determined by the capability module **240**. The user interface may display interactive elements that allow a user to request a workout plan, recommendations, or a current capability for an exercise or indicate that they have completed an exercise or request a new exercise in the workout plan (such as by swiping away a presented exercise). For example, a user may indicate, via a sliding interactive element of the user interface, a selected level of variance for a workout plan. The user may also enter information for their profile and whether they completed one or more exercises via the user interface. The user interface module **260** may update the profile of a user in the user profile datastore **270** when a user adds or changes their information for their profile, such as by adding new exercise goals, rating exercises, and the like.

[0043] The user interface module **260** may display exercises for a workout plan to a user via the user interface. For instance, a user may input, via the user interface, a request for a workout plan. The request may be associated with a level of variance selected by the user on a sliding interactive element, which allows a user to configure how much variation from other workout plans the user would like for the workout plan. The user interface module **260** sends the request to the workout plan module **200**, which determines a workout plan to send back to the user interface module **260** for display to the user. In some embodiments, the user interface module **260** may send one exercise (and a weight, number of repetitions, and number of sets) at a time from the workout plan for

display on the user interface, and upon receiving confirmation from the user via the user interface that the user performed the exercise, the user interface module **260** may update the user's exercise history in the user profile datastore **270**. Further, the user interface module **260** may send an indication to the workout plan module **200** to redetermine the workout plan for the user given that the user performed the exercise or may send an indication that the user has requested a different exercise.

[0044] The user interface module **260** may receive requests for a recommended weight, number of sets, or number of repetitions from a user via the user interface. In particular, the user interface may display a plurality of exercises for the user to choose from, and upon receiving a selection, the user interface module **260** may send an indication of the exercise to the recommendation module **220**. The user interface module **260** receives the recommendations from the recommendation module **220** and sends the recommendations for display via the user interface.

[0045] The user interface module **260** may also receive requests from a user for the user's capability for one or more exercises. The user interface module **260** may receive a request from the user interface for a current capability of the user for an exercise. The user interface module **260** sends the exercise to the capability module **240** to determine a capability of the user for the exercise. The capability of the user may be a weight for the exercise (e.g., the user can do one bench-press using 100 pounds) or a number of repetitions during a set (e.g., the user can do 32 pushups in one set). The user interface module **260** receives a capability of the user from the capability module **240** and displays the capability via the user interface.

Exercise Recommendation System Processes

[0046] FIGS. **3**, **4**, and **5** include flowchart illustrating processes (**300**, **400**, and **500**) performed by the exercise recommendation system **100**. Though reference is made to engines, modules, and models of the exercise recommendation system **100** for these processes, the processes can be used by other online systems or mobile applications.

[0047] FIG. **3** illustrates a process **300** for generating a workout plan, according to one embodiment. The user interface module **260** receives **310** a profile of a user and a level of variance selected by the user via a user interface, both of which the user interface module **260** sends to the workout plan module **200** and/or stores in the user profile datastore **270**. The workout plan module **200** inputs **320** the profile to the exercise selection model **210**, and the exercise selection model **210** is configured to rank a set of exercises for a user to perform based on the user's profile. The set of exercises may be specific to the user or may be standard for the exercise recommendation system **100**. The workout plan module **200** modifies **330** the ranking of exercises based on the profile and the level of variance. For instance, the workout plan module **200** may modify the ranking a greater amount for a first level of variance than for a second level of variance lower than the first level of variance. The workout plan module **200** generates **340** a workout plan for the user, where the workout plan comprises a threshold number of exercises selected based on the modified ranking.

[0048] It is appreciated that although FIG. **3** illustrates a number of interactions according to one embodiment, the precise interactions and/or order of interactions may vary in different embodiments. For example, in some embodiments, modifying the ranking comprises alerting scores of one or more exercise based on the user profile. In particular, the workout module **200** may reduce the scores of exercises that have low ratings from the user, that were recently performed by the user, that target muscle groups different from those specified in the user's exercise goals, that require equipment the user does not have, that conflict with physical ailments of the user, and the like. Further, the workout module **200** may increase the scores of exercises that target muscle groups specified by the user's exercise goals, that burn calories similar to a target set by the user, or the like. The workout module **200** may re-rank the exercises based on the altered scores and select exercises for the workout plan based on the new ranking.

[0049] FIG. **4** illustrates a process **400** for modifying a user interface to display a target weight. In particular, the recommendation module **220** accesses **410** a plurality of exercises pairs from the

exercise datastore **280**. Each exercise pair is labeled with performance statistics of users who have completed the exercises of the exercise pair. In some embodiments, each exercise pair may further be labeled based on muscle group capabilities, such as which muscle groups each exercise works. The training module **250** trains **420** a machine learning model (e.g., the weight recommendation model **230**) using the plurality of exercise pairs. The machine learning model is configured to determine a weight to recommend to a user for a first exercise based on performance statistics associated with one or more second exercises, where each second exercise is within an exercise pair with the first exercise. In some embodiments, the machine learning model is further configured to determine a number of repetitions to recommend to the user for the first exercise based on the performance statistics of the user associated with the one or more second exercises.

[0050] The recommendation module **220** retrieves **430** performance statistics of a target user from the user profile datastore **270**. The performance statistics may include a weight for each of one or more exercises previously performed by the target user, a weight of the user, a current muscle mass of the user, an average exercise time, a current capability of a user, and a number of repetitions for each of one or more exercises previously performed by the target user. The recommendation module **220** applies **440** the machine learning model to the performance statistics and a target exercise to determine a target weight to recommend to the target user for the target exercise. The recommendation module **220** modifies **450** the user interface displayed at a client device **120** of the target user to include the target weight determined by the machine learning model.

[0051] It is appreciated that although FIG. 4 illustrates a number of interactions according to one embodiment, the precise interactions and/or order of interactions may vary in different embodiments. For example, in some embodiments, responsive to receiving a request via the user interface for a target weight or target number of repetitions for an exercise, the recommendation module **220** applies the machine learning model to the performance statistics of the target user and the exercise to determine a target weight and/or target number of repetitions to recommend to the user for the exercise and modifies the user interface to include the target weight and/or target number of repetitions.

[0052] FIG. 5 illustrates a process **500** for determining a current capability of a user, according to one embodiment. The capability module **240** accesses **510** an exercise history for a user from the user profile datastore **270**. The exercise history comprises one or more exercises performed by the user and a capability of the user each time the user performed the exercise. In some embodiments, the exercise history may further comprise information about exercises performed within a recent time period and, for each exercise in the exercise history, a number of sets performed by the user, a number of repetitions performed by the user for each set, and a weight used by the user.

[0053] The capability module **240** partitions **520** the exercise history into a plurality of time periods, and for each time period, the capability module **240** computes **530** an aggregate capability for each of the one or more exercises based on the capabilities of the user in performing each exercise during the time period. The capability module **240** calculates **540** a moving average capability of the user for each exercise based on the aggregate capabilities. In some embodiments, the capability module **240** may apply exponential smoothing to the moving average capability. The capability module **240** determines **550** the current capability of the user for each exercise based on the moving average capability, where the current capability of the user for each exercise is discounted at least in part based on how recently the user performed the exercise.

[0054] It is appreciated that although FIG. 5 illustrates a number of interactions according to one embodiment, the precise interactions and/or order of interactions may vary in different embodiments. For example, in some embodiments, responsive to determining that a threshold amount of time has passed since the user has exercised, the capability module **240** may decay the current capability of the user based on the threshold amount of time that has passed and use the decayed current capability as the current capability for the user. In other embodiments, the capability module **240** may modify a user interface of a client device **120** associated with the user

to display the current capability responsive to receiving a request via the user interface for the current capability.

Summary

[0055] The foregoing description of the embodiments of the invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

[0056] Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as engines, without loss of generality. The described operations and their associated engines may be embodied in software, firmware, hardware, or any combinations thereof.

[0057] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software engines, alone or in combination with other devices. In one embodiment, a software engine is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

[0058] Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

[0059] Embodiments of the invention may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

[0060] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

Claims

1. A method for generating a workout plan, the method comprising: inputting, by a processor, a user profile to an exercise selection model to rank a set of exercises for a user to perform based on a history of exercises the user has performed, available gym equipment, and one or more exercise goals of the user; modifying, by the processor, the ranking of exercises based on a level of variance selected by the user, wherein the modification comprises: selecting a random set of exercises performed by the user within a threshold period of time based on the level of variance; and

adjusting rankings of the selected random set of exercises; applying, by the processor, a weight recommendation machine learning model to performance statistics of the user and a first exercise from the set of exercises, the weight recommendation machine learning model trained using supervised learning on training data including a plurality of exercise pairs each labeled with current capabilities of and weight recommendations for users that completed both exercises in a respective exercise pair; receiving, from the weight recommendation machine learning model, a current target weight to recommend to the user for the first exercise; and modifying, by the processor in real-time, a graphical user interface to display the current target weight and the first exercise.

2. The method of claim 1, wherein the user profile includes a recency of each exercise in the history of exercises the user has performed, exercise restrictions of the user, and one or more ratings for one or more exercises previously performed by the user.

3. The method of claim 1, wherein the level of variance indicates how much to vary one or more of each exercise, workout length, and focused muscle group in the ranking of exercises.

4. The method of claim 1, wherein the weight recommendation machine learning model is trained on training data including a plurality of exercise pairs each labeled with current capabilities of and weight recommendations for users that completed both exercises in a respective exercise pair.

5. The method of claim 4, wherein training the weight recommendation machine learning model using supervised learning creates a function that maps the exercise pairs to the respective current capabilities and weight recommendations.

6. The method of claim 1, further comprising: removing one or more exercises with low ratings by the user from the modified ranking.

7. The method of claim 1, further comprising: removing one or more exercises associated with equipment the user does not have from the modified ranking.

8. A non-transitory computer-readable storage medium comprising instructions executable by a processor, the instructions comprising: instructions for inputting, by the processor, a user profile to an exercise selection model to rank a set of exercises for a user to perform based on a history of exercises the user has performed, available gym equipment, and one or more exercise goals of the user; instructions for modifying, by the processor, the ranking of exercises based on a level of variance selected by the user, wherein the modification comprises: selecting a random set of exercises performed by the user within a threshold period of time based on the level of variance; and adjusting rankings of the selected random set of exercises; instructions for applying, by the processor, a weight recommendation machine learning model to performance statistics of the user and a first exercise from the set of exercises, the weight recommendation machine learning model trained using supervised learning on training data including a plurality of exercise pairs each labeled with current capabilities of and weight recommendations for users that completed both exercises in a respective exercise pair; instructions for receiving, from the weight recommendation machine learning model, a current target weight to recommend to the user for the first exercise; and instructions for modifying, by the processor in real-time, a graphical user interface to display the current target weight and the first exercise.

9. The non-transitory computer-readable storage medium of claim 8, wherein the user profile includes a recency of each exercise in the history of exercises the user has performed, exercise restrictions of the user, and one or more ratings for one or more exercises previously performed by the user.

10. The non-transitory computer-readable storage medium of claim 8, wherein the level of variance indicates how much to vary one or more of each exercise, workout length, and focused muscle group in the ranking of exercises.

11. The non-transitory computer-readable storage medium of claim 8, wherein the weight recommendation machine learning model is trained on training data including a plurality of exercise pairs each labeled with current capabilities of and weight recommendations for users that completed both exercises in a respective exercise pair.

- 12.** The non-transitory computer-readable storage medium of claim 11, wherein training the weight recommendation machine learning model using supervised learning creates a function that maps the exercise pairs to the respective current capabilities and weight recommendations.
- 13.** The non-transitory computer-readable storage medium of claim 8, the instructions further comprising: instructions for removing one or more exercises with low ratings by the user from the modified ranking.
- 14.** The non-transitory computer-readable storage medium of claim 8, the instructions further comprising: instructions for removing one or more exercises associated with equipment the user does not have from the modified ranking.
- 15.** A computer system comprising: a computer processor; and a non-transitory computer-readable storage medium storage instructions that when executed by the computer processor perform actions comprising: inputting, by the processor, a user profile to an exercise selection model to rank a set of exercises for a user to perform based on a history of exercises the user has performed, available gym equipment, and one or more exercise goals of the user; modifying, by the processor, the ranking of exercises based on a level of variance selected by the user, wherein the modification comprises: selecting a random set of exercises performed by the user within a threshold period of time based on the level of variance; and adjusting rankings of the selected random set of exercises; applying, by the processor, a weight recommendation machine learning model to performance statistics of the user and a first exercise from the set of exercises, the weight recommendation machine learning model trained using supervised learning on training data including a plurality of exercise pairs each labeled with current capabilities of and weight recommendations for users that completed both exercises in a respective exercise pair; receiving, from the weight recommendation machine learning model, a current target weight to recommend to the user for the first exercise; and modifying, by the processor in real-time, a graphical user interface to display the current target weight and the first exercise.
- 16.** The computer system of claim 15, wherein the user profile includes a recency of each exercise in the history of exercises the user has performed, exercise restrictions of the user, and one or more ratings for one or more exercises previously performed by the user.
- 17.** The computer system of claim 15, wherein the level of variance indicates how much to vary one or more of each exercise, workout length, and focused muscle group in the ranking of exercises.
- 18.** The computer system of claim 15, wherein the weight recommendation machine learning model is trained on training data including a plurality of exercise pairs each labeled with current capabilities of and weight recommendations for users that completed both exercises in a respective exercise pair.
- 19.** The computer system of claim 18, wherein training the weight recommendation machine learning model using supervised learning creates a function that maps the exercise pairs to the respective current capabilities and weight recommendations.
- 20.** The computer system of claim 15, the actions further comprising: removing one or more exercises with low ratings by the user from the modified ranking.
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